# Palindrome Checker - Three Approaches

Week 1 Practice - Problem 2 | Java

## Problem Summary

Check the same text using iterative two-pointer comparison, recursion, and character-array reversal; all three methods should agree.

## Java Solution

```
public class Main {
    static boolean isPalindromeIterative(String text) {
        int left = 0, right = text.length() - 1;
        while (left < right) {
            if (text.charAt(left) != text.charAt(right)) return false;
            left++; right--;
        }
        return true;
    }
    static boolean isPalindromeRecursive(String text) {
        return recursiveCheck(text, 0, text.length() - 1);
    }
    static boolean recursiveCheck(String text, int left, int right) {
        if (left >= right) return true;
        if (text.charAt(left) != text.charAt(right)) return false;
        return recursiveCheck(text, left + 1, right - 1);
    }
    static boolean isPalindromeArrayReversal(String text) {
        char[] chars = text.toCharArray();
        int left = 0, right = chars.length - 1;
        while (left < right) {
            char temp = chars[left]; chars[left] = chars[right];
            chars[right] = temp; left++; right--;
        }
        return text.equals(new String(chars));
    }
    static String label(boolean value) {
        return value ? "Palindrome" : "Not Palindrome";
    }
    static void check(String text) {
        System.out.println("Iterative: " + label(isPalindromeIterative(text))
            + " | Recursive: " + label(isPalindromeRecursive(text))
            + " | Array Reversal: " + label(isPalindromeArrayReversal(text)));
    }
    public static void main(String[] args) {
        check("madam"); check("hello");
    }
}
```

## Sample Output

```
Iterative: Palindrome | Recursive: Palindrome | Array Reversal: Palindrome
Iterative: Not Palindrome | Recursive: Not Palindrome | Array Reversal: Not Palindrome
```

## Complexity / Key Notes

Each method takes O(n) time. Iterative uses O(1) extra space, recursion uses O(n) call stack, and array reversal uses O(n) extra space. Comparison is case-sensitive and includes spaces.